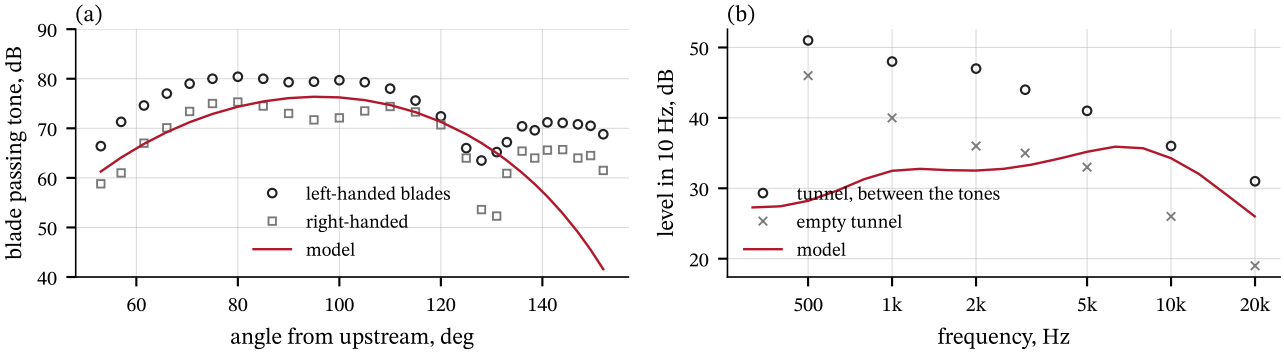
(a)
blade passing tone, dB
90
80
70
60
50
40
60
80
100
120
140
angle from upstream, deg
left-handed blades
right-handed
model
(b)
level in 10 Hz, dB
50
40
30
20
500
1k
2k
5k
10k
20k
frequency, Hz
tunnel, between the tones
empty tunnel
model